The Persistent Architecture of Relational Investment: Social Signatures in the NetHealth Study

Omar Lizardo

Department of Sociology, University of California, Los Angeles
NetHealth Research Collaboration

September 8, 2026

Abstract

How do individuals allocate cognitive, temporal, and emotional bandwidth across their personal networks? Using high-resolution digital call records paired with eight waves of longitudinal network and psychometric surveys from the NetHealth study ($N = 491$ egos, 498,237 communication events), this paper investigates the structure, temporal stability, and psychological foundations of **social signatures**—the ranked proportion of communication effort allocated across alters within discrete temporal windows. We show that the core signature persistence hypothesis holds across multiple timescales (academic years, semesters, quarters, months, and rolling weekly windows): intra-individual self-divergence (d_{self}) is uniformly and overwhelmingly smaller than inter-individual reference divergence (d_{ref}, $p < 10^{-12}$ across all resolutions). Parametric evaluations confirm that power-law models ($p(r) \sim r^{-\alpha}$, mean $R^2 = 0.935$) fit empirical signatures significantly better than exponential decay models, with ego-specific decay exponents (α) converging to an asymptotic value within 6 to 8 months of continuous observation. Going beyond previous work, we introduce three substantive theoretical expansions: (1) **Functional Support Grounding**: Linking communication ranks to multi-dimensional survey nominations ($N = 13,174$ dyads), we reveal that the steep drop in communication effort corresponds to sharp qualitative transitions from family-dominated instrumental and emotional support (Rank 1: 69.7% kin, 85.4% emotional support, 87.6% advice) to friend-dominated companionship at lower ranks; (2) **The Slot-Filling Dynamic**: We show that high signature stability persists despite massive alter turnover (averaging 80.1% across semesters), providing direct evidence that college students replenish relational slots with new alters rather than restructuring their overall allocation profile; and (3) **Psychological Determinants**: Using longitudinal mixed-effects and cross-sectional models, we show that Neuroticism strongly predicts steeper, hyper-concentrated signatures ($\beta = 0.098, p < 10^{-6}$), revealing that emotional vulnerability drives disproportionate reliance on core ties.

Keywords: Social Signatures, Egocentric Networks, NetHealth, Jensen-Shannon Divergence, Tie Strength, Slot-Filling Hypothesis, Personality Traits.

1 Introduction

A defining puzzle of human sociality is the tension between relational fluidity and structural constraint. Across the life course, individuals move through dynamic, evolving social worlds: entering new institutions, forming romantic partnerships, making new friends, and allowing older ties to lapse. Yet despite the continual turnover of interaction partners, personal communication behavior does not diffuse randomly across acquaintances, nor does it restructure arbitrarily when network composition changes. Instead, human relational investment is bounded by deep-seated cognitive and temporal constraints on sociality (Dunbar, 1992, 1998; Miritello et al., 2013). While individuals can recognize hundreds of faces and maintain casual awareness of extensive social circles, active personal networks are structured into concentric, hierarchical layers: an intimate support clique of 1 to 2 confidants, a sympathy group of approximately 5 close ties, an affinity group of 12 to 15 regular contacts, and expanding outer bands of declining emotional intensity and contact frequency (Dunbar, 2018; Roberts et al., 2009; Sutcliffe et al., 2012).

In a seminal contribution, Saramäki et al. (2014) operationalized this relational hierarchy through the concept of the **social signature**: the probability distribution of communication effort that an individual (ego) directs toward their contacts (alters), ordered from most-frequently to least-frequently contacted within a given observation window. Tracking young adults making the transition from high school to university using mobile call detail records, Saramäki et al. (2014) uncovered two foundational regularities: first, social signatures exhibit marked inter-individual variation, ranging from steep, hyper-concentrated profiles focused on a single alter to broad, egalitarian allocations; and second, an individual's social signature displays remarkable intra-individual persistence over time. Crucially, this persistence held even when personal networks underwent massive turnover, prompting Saramäki et al. (2014) to hypothesize a "slot-filling" dynamic in which individuals maintain stable cognitive slots and replenish them with new alters as old relationships fade.

Over the past decade, a rich multidisciplinary literature has expanded upon Saramäki and colleagues' initial formulation. Scholars have investigated social signatures across diverse communication channels, including text messaging (Heydari et al., 2018), online collaborative platforms and social networking sites (Li et al., 2018; Koltsova et al., 2021), and in-person sensor networks (Li and Bond, 2022; Adel et al., 2026). Physicists and network scientists have developed continuous resource allocation models (Tamarit et al., 2022) and generative micro-foundations rooted in the competition between cumulative advantage and random exploration (Iñiguez et al., 2023), while methodological work has sought to disentangle genuine tie strength heterogeneity from network size stability (Heydari et al., 2024; Heydari, 2024). Recent inquiries have even leveraged signature distinctiveness as a behavioral biometric capable of re-identifying individuals in anonymized data (Jia et al., 2025).

Despite these theoretical and computational advances, fundamental open questions remain at the heart of the social signatures framework. Most critically, because previous investigations have relied almost exclusively on commercial telecommunications metadata or passive digital traces, researchers have encountered a severe "content black box": commercial logs record the timing and frequency of calls or messages, but contain zero information regarding the functional support, emotional content, or qualitative nature of the underlying relationships. Does the steep mathematical drop-off between Rank 1 and lower-ranked alters correspond to qualitative distinctions in social support, such as emotional comfort, instrumental advice, companionship, or financial aid? Furthermore, the behavioral mechanisms governing signature stability remain empirically unverified: does signature persistence truly reflect active "slot-filling," or does it merely mirror the retention of core ties? What psychological dispositions and personality traits produce steep versus flat signatures in the first place? And over what observational timescales do empirical signatures converge to stable, asymptotic parameters?

This study addresses these fundamental questions by analyzing the **NetHealth Project**—a rich, longitudinal cohort study tracking an incoming class of undergraduate students at the University of Notre Dame across two full academic years (2015–2017). By pairing continuous, passive smartphone call records ($N = 491$ egos; 498,237 communication events) with an academic calendar, eight waves of sociocentric network surveys ($35,913$ alter nominations), complete alter-alter structural edge lists ($174,748$ ties), and longitudinal psychometric batteries (Big Five personality traits, CES-D depression, UCLA loneliness), we construct the most comprehensive, multidimensional empirical investigation of social signatures to date. We systematically evaluate signature persistence across multiple calendar and rolling timescales, test parametric power-law versus exponential scaling, establish the observation duration required for parameter convergence, ground communication ranks in multidimensional functional support dimensions, test the slot-filling dynamic under top-alter turnover, and estimate the psychological and structural drivers of signature morphology.

The remainder of this paper is organized as follows. Section 2 reviews the intellectual lineage and recent advances of the social signatures literature, identifying five critical theoretical and empirical frontiers addressed by our study. Section 3 details the NetHealth dataset, temporal window binning architectures, and mathematical formulations of social signatures and divergence metrics. Section 4 presents the empirical baseline results on persistence, power-law scaling, and parameter convergence. Section 5 develops four substantive expansions: functional support grounding, the slot-filling mechanism, multilevel panel models of signature divergence, and personality determinants of signature shape. Section 6 discusses the theoretical implications of our findings for evolutionary anthropology, sociology, and computational social science.

2 The Social Signatures Framework: Lineage, Advances, and Open Frontiers

The concept of the social signature sits at the intersection of evolutionary anthropology, structural sociology, and computational network science. In this section, we review the evolution of this framework over the past decade, synthesizing findings from mobile telecommunications, online interaction platforms, generative physical models, and methodological refinements. We then articulate five foundational frontiers left open in the prior literature that the NetHealth study is uniquely designed to address.

2.1 Foundational Origins: Cognitive Constraints and Persistent Skewness

The theoretical foundation of the social signatures framework is rooted in the social brain hypothesis (Dunbar, 1992, 1998, 2018), which posits that the evolution of large primate brains was driven by the cognitive demands of managing complex social relationships. In humans, neocortical processing capacity imposes an upper constraint on the number of personal ties an individual can actively maintain, a ceiling widely recognized as Dunbar's number (≈ 150). Within this outer boundary, egocentric personal networks are organized into a nested hierarchy of concentric layers or circles, where group sizes scale by a factor of approximately 3: an intimate support clique of 1 to 2 primary partners, a sympathy group of roughly 5 close ties, an affinity group of 12 to 15 core friends, and broader layers of 50 and 150 acquaintances (Roberts et al., 2009; Sutcliffe et al., 2012).

Crucially, maintaining social ties requires not only cognitive processing power but also the expenditure of finite temporal and emotional resources (Miritello et al., 2013). Because time is a non-renewable, zero-sum budget, individuals cannot allocate attention uniformly across their contacts; investing heavily in one relationship inevitably constrains the bandwidth available for others.

In a foundational paper, Saramäki et al. (2014) formalized this resource allocation process by introducing the **social signature**. Rather than treating personal networks as unweighted graphs or simple categorical layers, Saramäki and colleagues conceptualized an ego's relational environment as a continuous allocation profile. For an ego i maintaining contacts with K_i alters over an observation window w, alters are sorted in descending order of communication effort (measured by contact frequency or total call duration), $w_{i,(1),w} \geq w_{i,(2),w} \geq \cdots \geq w_{i,(K_i),w}$. The social signature is defined as the normalized vector of interaction proportions directed to each rank $r \in \{1, \ldots, K_i\}$:

$$p_{i,w}(r) = \frac{w_{i,(r),w}}{\sum_{k=1}^{K_i} w_{i,(k),w}}, \quad \text{such that} \quad \sum_{r=1}^{K_i} p_{i,w}(r) = 1. \tag{1}$$

To compare signatures across time or between different individuals, Saramäki et al. (2014) used the Jensen-Shannon Divergence (JSD) with zero-padding on-the-fly (Lin, 1991). For two signatures P and Q, zero-padded to common length $K = \max(|P|, |Q|)$, the distance is:

$$\mathrm{JSD}(P \parallel Q) = H(M) - \frac{1}{2}\left[H(P) + H(Q)\right], \tag{2}$$

where $M = \frac{1}{2}(P + Q)$ is the midpoint distribution and $H(\cdot)$ denotes the Shannon entropy in bits, $H(P) = -\sum_{r=1}^{K} p_r \log_2(p_r)$.

Tracking 24 high school graduates transitioning into university over 18 months, Saramäki et al. (2014) established two defining empirical patterns:

1. **Marked Inter-Individual Diversity**: Individuals possess distinct communication profiles.

While all signatures exhibit heavy-tailed skewness, some egos display hyper-concentrated distributions directing over 40% of their calls to Rank 1, whereas others distribute contact effort much more evenly across top alters.

2. **Intra-Individual Persistence Across Turmoil**: An individual's signature is remarkably stable over time. The self-divergence between an ego's signatures across consecutive observation periods (d_{self}) was significantly smaller than the reference divergence between random pairs of individuals (d_{ref}).

Most remarkably, this signature persistence endured despite massive turnover in network membership: between high school and university, participants replaced between 60% and 80% of their active contacts (Jaccard distance $1 - J \approx 0.60$–0.80). This striking paradox led Saramäki et al. (2014) to hypothesize the existence of a "slot-filling" mechanism: individuals do not adapt their overall allocation profile to the identities of their current alters; rather, they maintain pre-existing relational slots and dynamically slot newly acquired alters into vacated positions.

2.2 The Multichannel and Multimodal Frontier: Cross-Platform Consistency

Following Saramäki and colleagues' initial findings, a major research thrust examined whether social signatures are idiosyncratic to traditional voice telephony or reflect a universal organizing principle across modern communication channels.

First examining commercial mobile telecommunications at large scale, Heydari et al. (2018) analyzed voice call and Short Message Service (SMS) records from a European carrier spanning more than half a million egos. They confirmed that the signature persistence observed by Saramäki et al. (2014) in a small student cohort held across nationwide populations. Crucially, Heydari et al. (2018) showed that text messaging signatures also exhibited intra-individual persistence over time ($d_{\text{self}} < d_{\text{ref}}$). Furthermore, when comparing call and SMS signatures for the same ego, they found a moderate structural resemblance, even though the specific alters contacted via voice calls and text messages showed minimal overlap.

Broadening the inquiry to online environments, Li et al. (2018) investigated the dynamical evolution of social signatures among users of Facebook Wall posts and Wikipedia Talk pages across multiple consecutive intervals. Their empirical findings confirmed that social signatures remain stable in collaborative online networks despite continuous member churn. Extending the analytical framework, Li et al. (2018) introduced the concept of "structural signatures" based on topological embeddedness (the count of shared mutual friends between ego and alter in the global network). They showed that alters ranked by structural embeddedness produce rank-frequency profiles that mirror interaction-based social signatures, showing that behavioral communication intensity reflects positions within the wider community structure.

In a comprehensive multi-modal study, Li and Bond (2022) analyzed high-resolution data from the Copenhagen Networks Study (Stopczynski et al., 2014), tracking more than 700 university students across three distinct interaction modes: mobile phone calls, text messages, and Bluetooth-sensed in-person physical proximity. Li and Bond (2022) showed that social signatures were persistent across time and consistent across all three communication channels. However, they uncovered key modality-specific differences: communication networks mediated by voice calls and text messages were substantially more stable than in-person proximity networks, which exhibited higher turnover and greater susceptibility to circumstantial fluctuations. Drawing on media richness theory (Daft and Lengel, 1983), Li and Bond (2022) argued that individuals match the richness and synchronicity of specific communication channels to the ambiguity of relational tasks, choosing lean mediated channels for core ties while using face-to-face contact for broader social exploration.

More recently, Liu et al. (2024) evaluated multi-channel mobile communications using entropy evaluation methods and null models. Their analyses revealed an empirical U-shaped trade-off between calling and texting as an individual's total contact portfolio expands, alongside clear vitality ceilings: egos maintained active communication vitality up to approximately 10 contacts in voice calling, 100 in text messaging, and 20 in hybrid communication. Together, these studies confirm that social signatures represent an enduring, cross-modal behavioral pattern, but also indicate that different communication media capture distinct slices of an individual's total relational life.

2.3 Theoretical Formalisms and Universal Scaling Dynamics

Parallel to empirical extensions across platforms, theorists and statistical physicists sought to identify the mathematical laws and generative mechanisms that govern social signatures. A prominent debate concerned whether personal networks are best understood as discrete, stepwise layers (the classical Dunbar circles) or as smooth, continuous distributions.

Bridging this divide, Tamarit et al. (2022) formulated a statistical-mechanical model of resource allocation based on the principle of maximum entropy. Rather than categorizing ties into a small number of arbitrary discrete bins where the cost of interaction is assumed to be uniform within each layer, Tamarit and colleagues treated relational maintenance costs as continuous variables. Their continuum model showed that discrete Dunbar circles and continuous social signatures are mathematically unified manifestations of a single resource allocation process under finite cognitive capacity. Under this formulation, they predicted and empirically identified a universal scaling parameter, $\eta \approx 6$, representing the ratio of relationship costs between adjacent levels. Evaluating datasets spanning mobile phone logs, face-to-face contacts, and Facebook interactions, Tamarit et al. (2022) showed that the distribution of estimated parameters across individuals clustered tightly around this universal value.

Simultaneously, Iñiguez et al. (2023) tackled the generative micro-foundations of tie strength heterogeneity. Analyzing personal networks across millions of individuals in four massive datasets (mobile calls, text messages, Twitter mentions, and Wikipedia talk pages), they showed universality in the distributions of tie strengths and their individual-level variations. To explain this universality, Iñiguez et al. (2023) introduced a microscopic model of egocentric network evolution driven by a dynamic competition between two fundamental behavioral mechanisms:

1. **Cumulative Advantage (Preferential Tie Reinforcement)**: An ego directs communica-

tion to existing alters with probability proportional to past interactions, parameterized by an alter-preferentiality exponent, w_{ir}^{β}.

2. **Random Choice (Exploratory Communication)**: An ego explores new ties or contacts existing alters at random, independent of past interaction history.

The balance between these forces is captured by the alter-preferentiality parameter $\beta \in [0, \infty)$. When $\beta = 0$, communication is allocated uniformly across alters; as β increases, cumulative advantage concentrates communication into a narrow elite of alters. Iñiguez et al. (2023) showed that individual variation in β fully accounts for empirical differences in social signature steepness, proving that heavy-tailed signatures reflect a universal balancing act between exploiting core relationships and exploring peripheral connections.

Extending this generative framework to offline physical settings, Adel et al. (2026) analyzed face-to-face wearable sensor logs and spatial colocation traces alongside online communication networks. Their findings revealed that while spatial colocation (physical proximity without active engagement) is a noisy proxy that fails to exhibit social signature structure, active face-to-face interactions conform precisely to the preferentiality dynamics identified by Iñiguez et al. (2023). Crucially, Adel et al. (2026) showed that empirical personal networks operate under a bounded communication bandwidth, identifying a universal scaling relationship between tie preferentiality (β) and bandwidth capacity that holds across both online and offline social systems.

2.4 Methodological Refinements: Disentangling Network Size from Tie Strength Heterogeneity

As the empirical literature expanded, methodologists noted an important vulnerability in how social signature persistence had historically been measured. In a rigorous examination, Heydari et al. (2024) and Heydari (2024) pointed out that an ego's social signature contains two distinct structural features: its length (the personal network size or degree, k) and the internal distribution of interaction weights across those ranks (tie strength heterogeneity).

Standard evaluations of signature persistence calculate the Jensen-Shannon Divergence or Euclidean distance between zero-padded signature vectors. However, Heydari et al. (2024) showed that JSD distance is mathematically sensitive to differences in signature length: two signatures of similar length will mechanically yield a lower divergence than two signatures of disparate lengths, regardless of their internal curvature. Because an individual's network degree tends to be relatively stable across consecutive observation windows, the self-divergence (d_{self}) is systematically calculated between signatures of similar length. Conversely, the reference divergence (d_{ref}) compares random pairs of individuals in the population, who often exhibit substantial degree disparities. Consequently, standard JSD comparisons risk conflating the persistence of network size with the persistence of communication allocation strategy.

To resolve this confounding, Heydari et al. (2024) evaluated degree-insensitive metrics of tie strength heterogeneity, including the Gini coefficient distance:

$$\Delta G(G_1, G_2) = |G_1 - G_2|, \quad \text{where} \quad G = \frac{2}{k} \frac{\sum_{r=1}^{k} r \cdot w_r}{\sum_{r=1}^{k} w_r} - \frac{k+1}{k}, \tag{3}$$

and the normalized alter-preferentiality distance derived from Iñiguez and colleagues' generative model:

$$\Delta\beta(\beta_1, \beta_2) = \frac{|\beta_1 - \beta_2|}{\beta_1 + \beta_2}. \tag{4}$$

Applying these degree-invariant measures to large-scale mobile phone datasets, Heydari et al. (2024) proved that the persistence of personal network structure cannot be attributed to degree stability alone. Even after strictly controlling for degree differences, intra-individual distances remained significantly smaller than inter-individual reference distances. This confirmed that tie strength heterogeneity is an authentic, independent behavioral trait of egos: individuals possess an enduring personal allocation style (whether concentrated or egalitarian) that persists independently of how many contacts they maintain.

2.5 Longitudinal Drift, Biometric Uniqueness, and Privacy

Finally, two recent developments have examined the longitudinal boundaries of signature persistence and its implications for individual privacy.

Investigating the temporal stability of social signatures over extended observation horizons, Koltsova et al. (2021) tracked private messaging on the Russian social networking site VKontakte (VK) across 18 consecutive months. While replicating the baseline finding that short-term self-distances were smaller than inter-individual reference distances ($d_{\text{self}} < d_{\text{ref}}$), Koltsova et al. (2021) made a critical new discovery: intra-individual signature changes across longer spans (e.g., between Months 1–6 and Months 13–18) were statistically significant via Kolmogorov-Smirnov tests. They argued that social signatures are not rigid, immutable constants; rather, they undergo subtle developmental drift as individuals experience shifting life circumstances, suggesting that signature stability is bounded by long-term life transitions.

Conversely, the very uniqueness and temporal persistence of personal allocation profiles has raised significant data privacy concerns. Showing this potential for behavioral re-identification, Jia et al. (2025) examined how multidimensional social signatures constructed from interaction timing, contact volume, and communication ranks can de-anonymize users in supposedly anonymous datasets. Using low-sensitivity interaction metadata from email, call logs, and app engagement records, Jia et al. (2025) achieved user re-identification accuracies of up to 87%. Their findings establish that an individual's communication signature is so distinct and persistent that it operates as a behavioral biometric fingerprint, challenging conventional assumptions regarding the safety of anonymized network data under privacy regulations such as the GDPR and CCPA.

2.6 Critical Open Questions and Frontiers Addressed by the NetHealth Study

Despite the substantial progress achieved across these studies, the literature has reached an empirical impasse defined by five major open questions that existing commercial datasets cannot answer:

1. **The Content Black Box: Grounding Mathematical Ranks in Functional Social Support**: Commercial telecommunications records (Saramäki et al., 2014; Heydari et al., 2018; Iñiguez et al., 2023), online messaging logs (Li et al., 2018; Koltsova et al., 2021), and Bluetooth sensor traces (Li and Bond, 2022) are completely devoid of relational content. Consequently, researchers have been unable to determine what communication ranks actually mean in human terms. Does the steep drop in communication effort between Rank 1 and lower-ranked alters reflect mere quantitative differences in contact volume, or does it mark qualitative transitions between distinct functional support dimensions (e.g., emotional confiding, instrumental advice, social companionship, financial aid, or kinship versus peer friendship)? While Koltsova et al. (2021) examined a 1-to-9 subjective closeness rating on a small sample of 39 users, no study has linked large-scale communication ranks to multidimensional sociometric support taxonomies.

   The NetHealth Resolution: We cross-reference $N = 13,174$ call-ranked dyads with eight waves of longitudinal sociocentric surveys recording multidimensional social support (emotional comfort, advice, companionship, financial assistance) and tie categories (family kin versus peer friends). We show that communication rank tracks sharp, qualitative functional divisions across classic Dunbar tiers.

2. **Direct Empirical Verification of the Slot-Filling Mechanism**: While Saramäki et al. (2014) and Heydari (2024) hypothesized that signature stability persists because individuals slot new alters into pre-existing relational positions, they lacked the alter-level tracking and longitudinal survey depth required to verify this mechanism. Specifically, does an individual's signature remain stable because their core ties (Rank 1 and Rank 2) never change, or does stability truly endure when an individual experiences complete turnover of their primary confidant?

   The NetHealth Resolution: Leveraging continuous passive phone logs across eight academic semesters, we observe an average semester alter turnover of 80.1%. We explicitly separate transitions where an ego retains their primary alter (Rank 1) from transitions where the primary alter is completely replaced. We show that even under complete turnover of the top confidant, signature divergence shifts only marginally (from 0.043 to 0.078), providing direct empirical confirmation of the slot-filling hypothesis.

3. **The Psychological Foundations of Signature Curvature**: Theoretical models of personal network evolution (Dunbar, 2018; Iñiguez et al., 2023; Adel et al., 2026) assume that individual differences in alter-preferentiality (β) and signature decay exponents (α) reflect internal cognitive capacities and psychological dispositions. Yet, commercial telecommunications providers have no psychometric data on their subscribers, and university sensor studies (Li and

Bond, 2022) did not model personality predictors of signature steepness. Beyond preliminary exploratory correlations (Centellegher et al., 2017), no study has rigorously estimated how validated personality dimensions govern the curvature of social signatures.

The NetHealth Resolution: We link objective communication decay exponents (α_i) to longitudinal psychometric batteries measuring the Big Five personality traits (Extraversion, Agreeableness, Conscientiousness, Neuroticism, Openness), CES-D depression, and UCLA loneliness scores. Through regression modeling, we show that Neuroticism strongly predicts steeper, hyper-concentrated signatures ($\beta = 0.098, p < 10^{-6}$), revealing that emotional vulnerability drives disproportionate reliance on a narrow core of alters.

4. **Multi-Timescale Hierarchy and Parameter Burn-In Convergence**: Prior empirical investigations evaluated social signatures at arbitrary, isolated temporal resolutions: Li and Bond (2022) examined 1-month intervals, Heydari et al. (2018) used 6-month windows, and Saramäki et al. (2014) examined 9-month terms. However, no study has evaluated how signature stability behaves across a nested hierarchy of timescales within the same population. More crucially, prior literature offers no guidance on parameter burn-in: how long must an individual be continuously observed before their estimated signature decay parameter α converges to its true asymptotic value?

   The NetHealth Resolution: We evaluate social signatures across eight temporal window architectures spanning calendar definitions (academic years, semesters, quarters, months) and rolling weekly intervals (3-week rolling, 2-week discrete, 1-week discrete). Furthermore, by tracking cumulative observation intervals from 2 to 24 months, we establish an empirical burn-in threshold showing that personal signature parameters stabilize within 6 to 8 months of continuous observation.

5. **Sociocentric Structural Embedding and Multilevel Panel Dynamics**: Previous studies have modeled egocentric communication signatures in isolation from the structural connectivity of the surrounding network. Yet human communication is embedded within broader community structures, residence halls, and friendship groups (Li et al., 2018). How do alter turnover, egonet transitivity (clustering), and shifts in overall communication volume jointly govern signature stability over time?

   The NetHealth Resolution: NetHealth pairs continuous individual phone logs with 174,748 alter-alter structural ties connecting peers within each respondent's personal egocentric network across survey waves. We estimate multilevel linear mixed-effects models predicting signature self-divergence (d_{self}) across consecutive semesters as a joint function of dyadic Jaccard turnover, triadic clustering, and communication volume shifts, providing the first structural panel evaluation of signature divergence.

3 Data and Methodology

3.1 The NetHealth Dataset

The NetHealth study recruited an entire incoming cohort of first-year undergraduate students at the University of Notre Dame beginning in Fall 2015. Participants consented to continuous digital sensing via their smartphones and Fitbit devices, alongside dense sociocentric network and psychological surveys administered at regular intervals across their college careers.

Our analytical design links five primary data streams:

1. **Continuous Communication Logs**: 498,237 outgoing voice call events collected from participant iOS devices spanning June 30, 2015 through July 2, 2017 across 491 unique egos.

2. **Academic Calendar**: Comprehensive institutional records documenting semester start and end dates, examination periods, orientation, and vacation breaks.

3. **Sociocentric Network Surveys**: 35,913 alter nominations gathered across 8 longitudinal survey waves recording tie categories (family kin vs. peer friends), perceived closeness, interpersonal trust, and multidimensional social support.

4. **Alter-Alter Edge Lists**: 174,748 structural ties connecting peers within each respondent's personal egocentric network across survey waves.

5. **Basic Psychometric Batteries**: Longitudinal survey instruments measuring the Big Five personality dimensions (Extraversion, Agreeableness, Conscientiousness, Neuroticism, Openness), the CES-D Depression Scale, and the UCLA Loneliness Scale.

3.2 Temporal Window Binning and Filtering

To establish whether social signatures persist across differing temporal resolutions, we construct two complementary binning architectures:

- **Calendar-Based Windows** (July 1, 2015 – June 30, 2017):
  - *Academic Years* ($W = 2$): July 2015 – June 2016; July 2016 – June 2017.
  - *Semesters* ($W = 4$): Fall 2015, Spring 2016, Fall 2016, Spring 2017.
  - *Quarters* ($W = 8$): July–September, October–December, January–March, April–June.
  - *Months* ($W = 24$): 24 consecutive calendar months.
- **Week-Based Windows** (July 6, 2015 – July 2, 2017; 104 Monday–Sunday weeks):
  - *3-Week Rolling Windows* ($W = 102$): 3-week moving windows in 1-week increments.
  - *2-Week Discrete Windows* ($W = 52$): Non-overlapping 14-day intervals.
  - *1-Week Discrete Windows* ($W = 104$): 7-day intervals.

To ensure that estimated signatures reflect meaningful personal communication structures, egos are required to meet two standard inclusion criteria: (a) a minimum of 2 active alters in every window ($k_{i,w} \geq 2$), and (b) an average call volume exceeding 10 calls per window. When comparing across calendar definitions, we evaluate the common intersection cohort ($N = 147$ egos).

Table 1: NetHealth Cohort Summary Across Temporal Window Definitions

Window Scheme	**Window Span**	**Windows (W)**	**Eligibility Criteria**	**Eligible Egos (N)**
Academic Years	Jul 2015 – Jun 2017	2	≥ 2 alters, > 10 calls	388
Semesters	Jul 2015 – Jun 2017	4	≥ 2 alters, > 10 calls	290
Quarters	Jul 2015 – Jun 2017	8	≥ 2 alters, > 10 calls	227
Months	Jul 2015 – Jun 2017	24	≥ 2 alters, > 10 calls	147
Common Calendar Cohort	Jul 2015 – Jun 2017	–	All 4 calendar schemes	147
3-Week Rolling	Jul 2015 – Jul 2017	102	≥ 2 alters, > 10 calls	88
2-Week Discrete	Jul 2015 – Jul 2017	52	≥ 2 alters, > 10 calls	65
1-Week Discrete	Jul 2015 – Jul 2017	104	≥ 2 alters, > 10 calls	14

3.3 Mathematical Formulation of Social Signatures and Jensen-Shannon Divergence

For ego i in temporal window w, communication weights $w_{i,j,w}$ (measured as total outgoing voice calls) to alters $j = 1, \ldots, k_{i,w}$ are sorted in descending order:

$$w_{i,(1),w} \geq w_{i,(2),w} \geq \cdots \geq w_{i,(k),w} \tag{5}$$

The social signature is defined as the normalized vector of interaction proportions:

$$p_{i,w}(r) = \frac{w_{i,(r),w}}{W_{i,w}}, \quad \text{where } W_{i,w} = \sum_{j=1}^{k_{i,w}} w_{i,j,w} \quad \text{and} \quad \sum_{r=1}^{k_{i,w}} p_{i,w}(r) = 1 \tag{6}$$

where $r \in \{1, 2, \ldots, k_{i,w}\}$ represents the alter rank.

To quantify the similarity or difference between two probability vectors $P = \{p_r\}_{r=1}^{k_1}$ and $Q = \{q_r\}_{r=1}^{k_2}$ of potentially unequal lengths $k_1 \neq k_2$, we append zeros to the shorter distribution up to length $K = \max(k_1, k_2)$ on-the-fly and compute the **Jensen-Shannon Divergence** (JSD; Lin, 1991):

$$\text{JSD}(P \parallel Q) = H(M) - \frac{1}{2}\left[H(P) + H(Q)\right] \tag{7}$$

where $M = \frac{1}{2}(P + Q)$ is the mixture distribution, and $H(P)$ denotes Shannon entropy measured in bits:

$$H(P) = -\sum_{r=1}^{K} p_r \log_2(p_r) \quad (\text{with } 0 \log_2 0 \equiv 0) \tag{8}$$

JSD is symmetric ($\text{JSD}(P \parallel Q) = \text{JSD}(Q \parallel P)$), non-negative, and strictly bounded between 0 (identical distributions) and 1 bit (disjoint distributions).

For evaluating consistency across multiple observation windows simultaneously, we also compute the generalized n-ary Jensen-Shannon divergence across a set of n distributions $\{P_1, P_2, \ldots, P_n\}$:

$$\text{JSD}(P_1, \ldots, P_n) = H\left(\frac{1}{n}\sum_{m=1}^{n} P_m\right) - \frac{1}{n}\sum_{m=1}^{n} H(P_m) \tag{9}$$

We define two core divergence metrics:

1. **Self-Divergence (d_{self}):** The mean pairwise JSD between consecutive temporal windows for the same ego across observation periods:

$$d_{\text{self}}(i) = \frac{1}{W-1}\sum_{w=1}^{W-1} \text{JSD}(P_{i,w} \parallel P_{i,w+1}) \tag{10}$$

2. **Reference-Divergence (d_{ref}):** The divergence between the social signatures of distinct individuals observed within the same temporal window:

$$\text{By Ego: } d_{\text{ref}}(i) = \frac{1}{W}\sum_{w=1}^{W}\left[\frac{1}{n_w - 1}\sum_{j \neq i} \text{JSD}(P_{i,w} \parallel P_{j,w})\right] \tag{11}$$

$$\text{All Pairs: } \bar{d}_{\text{ref}} = \frac{1}{\binom{n}{2}}\sum_{i<j}\left[\frac{1}{W}\sum_{w=1}^{W} \text{JSD}(P_{i,w} \parallel P_{j,w})\right] \tag{12}$$

If personal social signatures exhibit persistent idiosyncratic stability over time, an individual's self-divergence must be systematically and significantly smaller than the reference divergence observed between random peers ($d_{\text{self}} < d_{\text{ref}}$).

4 Empirical Results: Persistence and Parametric Architecture

4.1 Empirical Social Signatures

Figure 1 displays empirical social signatures averaged across egos for ranks 1 through 15 across temporal window definitions. Across all timescales, communication effort exhibits pronounced skewness:

- The single top-ranked alter ($r = 1$) commands **35% to 45%** of an ego's total communication bandwidth.
- The second-ranked alter ($r = 2$) captures **18% to 22%**.
- The top 3 alters collectively account for over **70%** of total outgoing interaction volume.
- Beyond rank 5, communication drops below 4% per alter.

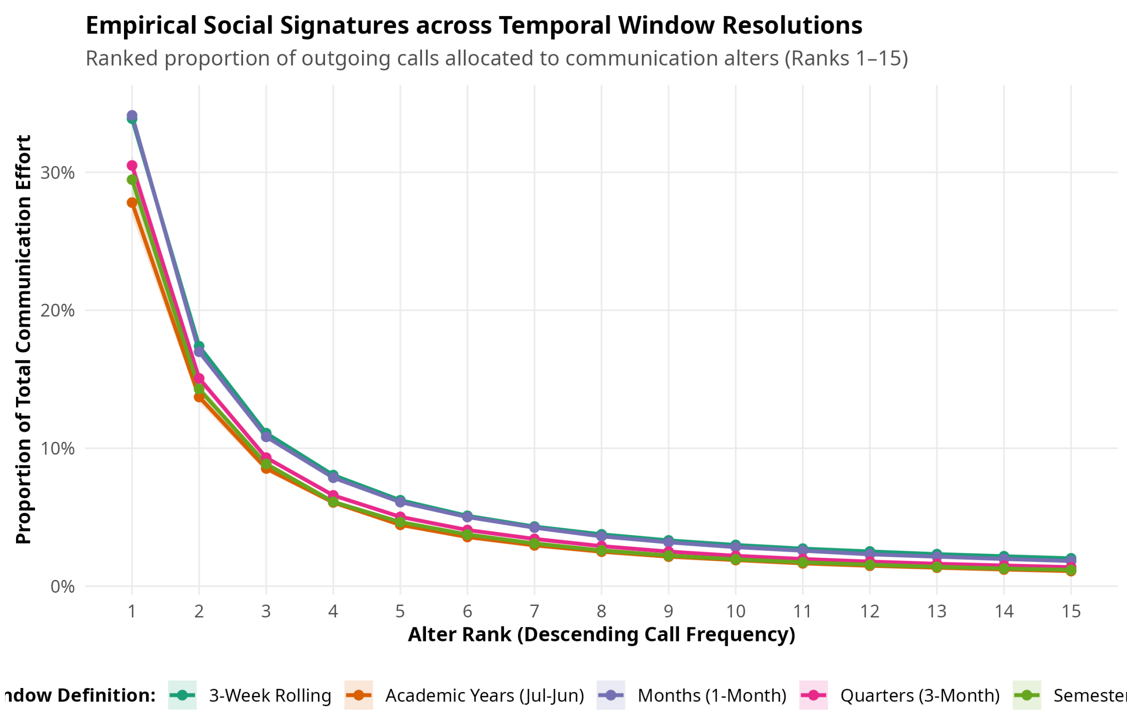

Figure 1: Empirical Social Signatures across Temporal Window Resolutions (Ranks 1–15). Ribbons denote 95% confidence intervals around the mean interaction proportion.

4.2 Persistent Individual Differences Across Timescales

Table 2 reports the formal statistical comparison between intra-individual self-divergence (d_{self}) and inter-individual reference divergence (d_{ref}).

Table 2: Statistical Tests of Social Signature Persistence Across Window Schemes

Scheme	Egos (N)	Mean d_{self} (SD)	Mean d_{ref} (SD)	Difference	Wilcoxon V	p-value
Academic Years	388	0.0633 (0.0642)	0.1129 (0.0322)	−0.0496	6,690.0	$< 10^{-12}$
Semesters	290	0.0614 (0.0423)	0.1184 (0.0304)	−0.0570	1,136.0	$< 10^{-12}$
Quarters	227	0.0626 (0.0311)	0.1293 (0.0299)	−0.0667	50.0	$< 10^{-12}$
Months	147	0.0877 (0.0305)	0.1494 (0.0248)	−0.0617	14.0	$< 10^{-12}$
Common Cohort (Sem)	147	0.0458 (0.0244)	0.1057 (0.0250)	−0.0599	30.0	$< 10^{-12}$
Common Cohort (Mo)	147	0.0877 (0.0305)	0.1494 (0.0248)	−0.0617	14.0	$< 10^{-12}$
3-Week Rolling	88	0.0495 (0.0147)	0.1411 (0.0205)	−0.0916	0.0	$< 10^{-12}$
2-Week Discrete	65	0.0992 (0.0216)	0.1439 (0.0167)	−0.0447	0.0	$< 10^{-12}$

Across every temporal resolution examined, intra-individual self-divergence is significantly lower than inter-individual reference divergence ($p < 10^{-12}$, Wilcoxon signed-rank test; Cohen's $d > 1.4$). In semester windows, mean self-divergence is 0.0614, compared to a reference divergence of 0.1184 ($V = 1136, p = 1.27 \times 10^{-44}$). Even at fine-grained weekly scales (3-week rolling windows), an individual's signature remains exceptionally consistent across time ($d_{\text{self}} = 0.0495$) relative to the broader population ($d_{\text{ref}} = 0.1411, p = 1.90 \times 10^{-16}$). As shown in Figure 2, the distributions of self and reference divergence exhibit minimal overlap, confirming that personal communication signatures reflect enduring individual traits.

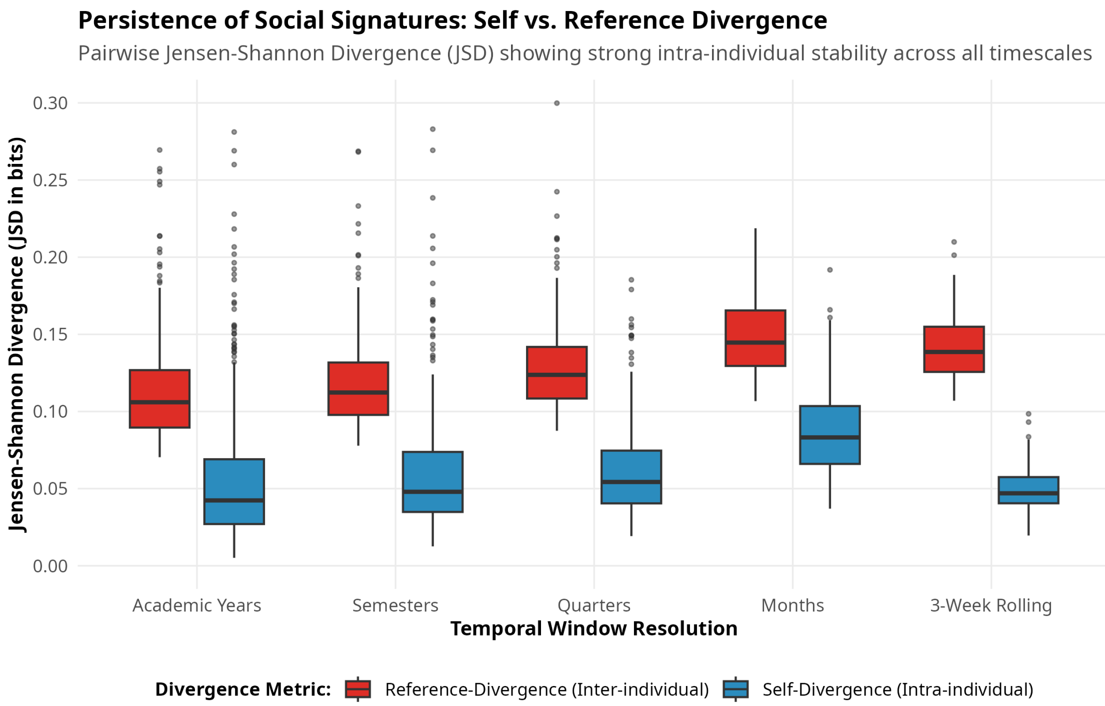

Figure 2: Persistence of Social Signatures: Self-Divergence vs. Reference-Divergence across temporal resolutions. Boxplots show the distribution of individual ego means.

4.3 Parametric Form: Power-Law vs. Exponential Decay

We evaluated two competing functional specifications for the signature curve:

$$\text{Power-Law: } p(r) = c \cdot r^{-\alpha} \iff \log p(r) = \log c - \alpha \log r \tag{13}$$

$$\text{Exponential: } p(r) = c \cdot e^{-\beta r} \iff \log p(r) = \log c - \beta r \tag{14}$$

Table 3: Parametric Model Evaluation Across Window Resolutions

Scheme	Models (N)	Power-Law α (SD)	Power-Law R^2	Exp β	Exp R^2	% PL Preferred
Semesters	1,158	1.20 (0.26)	0.935	0.10	0.746	97.0%
Months	3,496	1.06 (0.37)	0.883	0.23	0.771	83.8%

As summarized in Table 3 and Figure 3, the power-law model decisively outperforms the exponential specification across both semester and monthly timescales. In semester windows, the power-law specification is preferred by Akaike Information Criterion (AIC) in **97.0% of models**, yielding an average R^2 of 0.935 (compared to 0.746 for exponential decay). The power-law decay exponent α averages 1.22 (SD = 0.25), reflecting a heavy-tailed allocation profile.

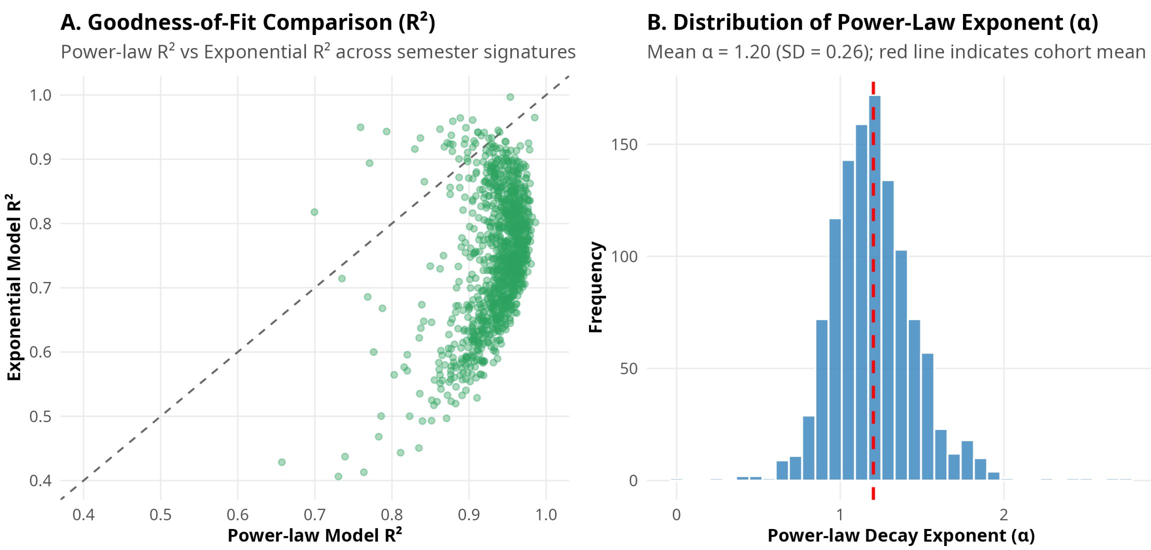

Figure 3: Power-Law vs. Exponential Model Fits. (A) Scatterplot of R^2 values comparing power-law against exponential fits across semester signatures. (B) Empirical distribution of power-law decay exponent (α).

4.4 Longitudinal Parameter Convergence and Burn-In

To establish how long an individual must be observed before their estimated signature parameters stabilize, we tracked the common cohort across cumulative observation lengths from 2 to 24 months (Figure 4).

- At 2 months: Mean $\alpha = 1.16$, with a mean absolute error of 0.188 relative to the 24-month asymptotic value.
- At 4 months: Mean absolute error declines to 0.141 ($R^2 = 0.934$).
- At 6 to 8 months: Mean absolute error converges below 0.10 ($R^2 = 0.945$).

These findings show that **6 months of continuous digital observation** provides an optimal burn-in threshold for reliably characterizing personal social signatures in young adult populations.

5 Theoretical Expansions

5.1 Functional Grounding of Communication Ranks in Social Support Dimensions

While prior literature has treated communication ranks as purely behavioral tallies, we linked $N = 13,174$ call-ranked dyads to longitudinal network surveys recording relationship categories and support functions across six theoretical Dunbar tiers (Table 4 and Figure 5).

The findings illuminate the qualitative architecture underlying social signatures:

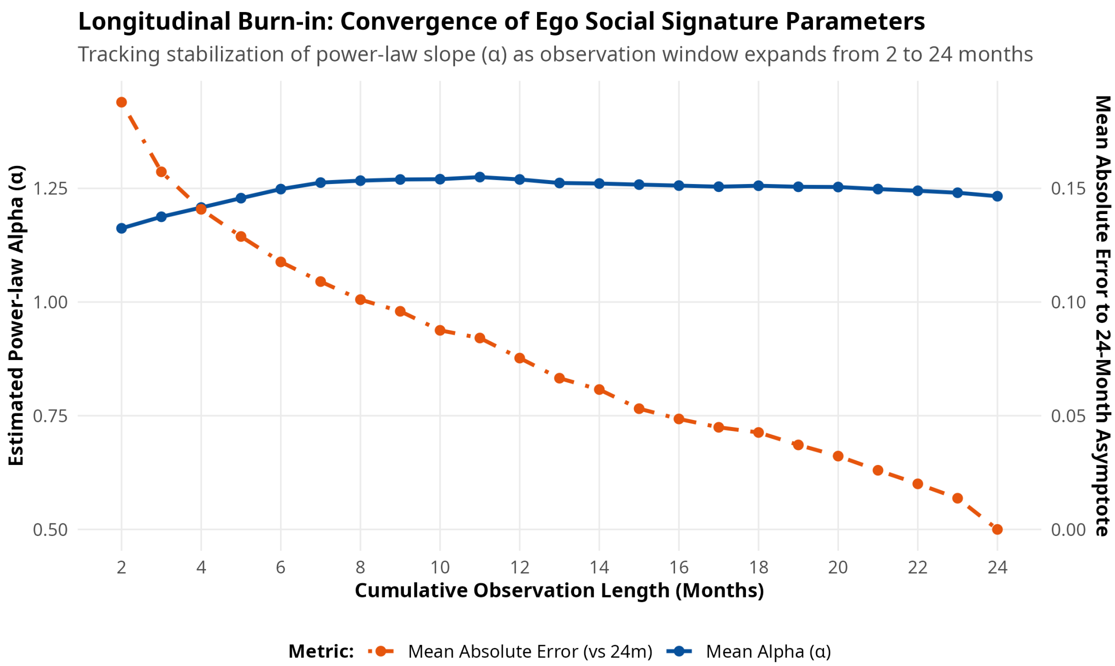

Figure 4: Parameter Burn-In Convergence Over 24 Months. Left axis shows mean estimated power-law α; right axis shows mean absolute error relative to the 24-month asymptotic parameter value.

Table 4: Relational Composition and Support Functions Across Signature Rank Tiers ($N = 13,174$ Dyads)

Rank Tier	Ties (N)	% Family	% Friend	% Emotional	% Advice	% Companionship	% Financial
Rank 1 (Core)	950	69.7%	16.9%	85.4%	87.6%	80.6%	64.1%
Ranks 2–3	1,713	62.2%	34.0%	77.1%	80.1%	79.8%	45.1%
Ranks 4–5	1,426	41.4%	56.9%	70.0%	73.0%	81.9%	24.2%
Ranks 6–10	2,743	24.2%	74.4%	60.7%	64.3%	78.2%	12.4%
Ranks 11–20	3,011	14.3%	84.5%	48.1%	53.6%	74.4%	7.1%
Ranks > 20	3,331	9.2%	87.8%	39.1%	44.0%	69.5%	4.4%

1. **The Kinship Core**: Rank 1 is heavily dominated by family members (69.7% kin, primarily parents and romantic partners), who provide the structural foundation of instrumental, advice, and financial support (64.1% financial assistance, 87.6% advice, 85.4% emotional support).

2. **The Friendship Transition**: A sharp structural crossover occurs between Ranks 3 and 5. At Ranks 2–3, family members still constitute 62.2% of ties; by Ranks 4–5, peer friends constitute the majority (56.9%), rising to 87.8% beyond Rank 20.

3. **Differentiated Support Gradient**: Emotional support (85.4% at Rank 1) and advice support (87.6% at Rank 1) fall monotonically across tiers (dropping to 39.1% and 44.0% for outer ties). In sharp contrast, companionship remains elevated across all tiers (80.6% at Rank 1, 81.9% at Ranks 4–5, and 69.5% beyond Rank 20), showing that peripheral communication ties serve primary social leisure and recreational functions.

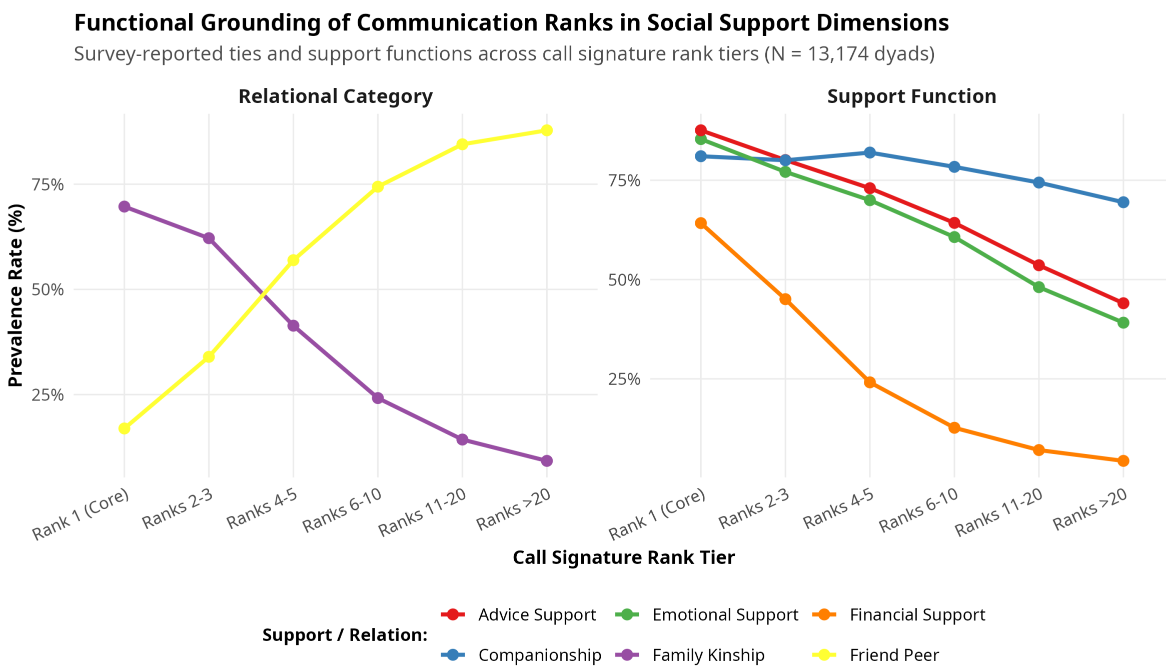

Figure 5: Functional Support Dimensions Across Signature Rank Tiers. Shows prevalence rates of relational categories (left panel) and substantive support provisions (right panel).

5.2 The "Slot-Filling" Dynamic: Alter Turnover vs. Signature Stability

A central theoretical puzzle in network science is how an ego's signature can remain stable despite frequent alter replacement. We calculated the dyadic Jaccard turnover between consecutive semesters:

$$\text{Turnover}_{i,w} = 1 - \frac{|A_{i,w} \cap A_{i,w+1}|}{|A_{i,w} \cup A_{i,w+1}|} \tag{15}$$

where $A_{i,w}$ denotes the set of active alters for ego i in window w.

Remarkably, **alter turnover between consecutive semesters averages 80.1% (SD = 0.076)**. Despite replacing four out of five alters every six months, egos maintain an average self-divergence of only 0.0614. In 59.1% of consecutive semester intervals, the single top-ranked alter was retained; even when the top alter was replaced, the overall self-divergence increased only modestly (from 0.0431 to 0.0785, Figure 6).

This pattern provides direct empirical confirmation of the **slot-filling hypothesis**: an individual maintains a predefined cognitive template for relationship allocation, seamlessly slotting new social entrants into vacant relational positions without perturbing their structural allocation curve.

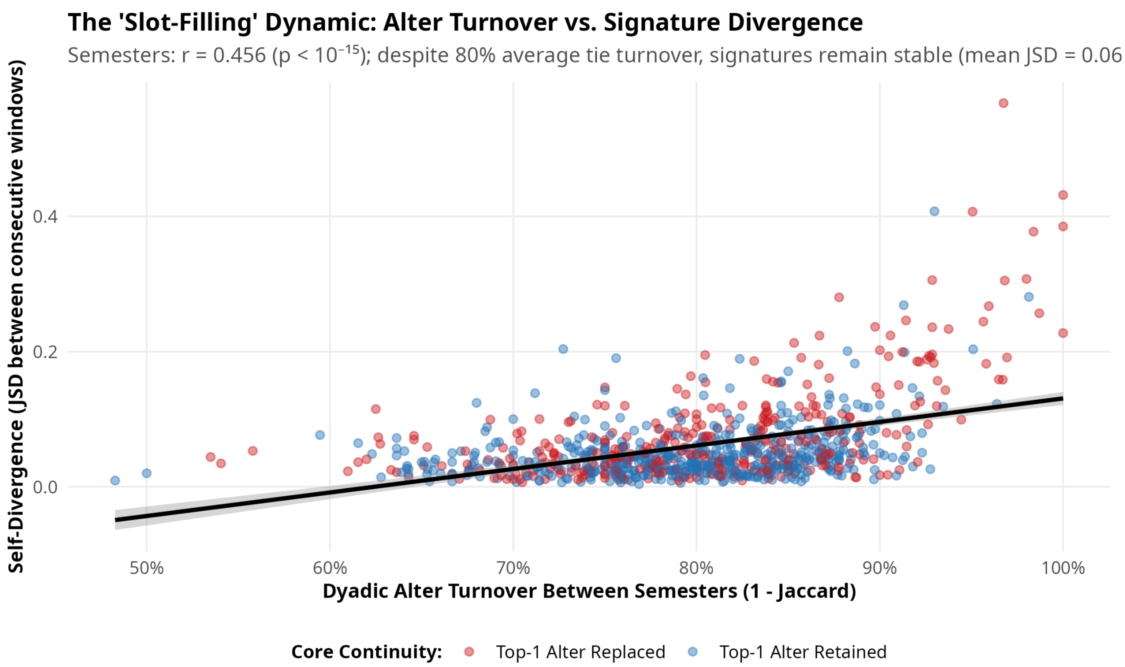

Figure 6: The 'Slot-Filling' Dynamic: Alter Turnover vs. Signature Divergence ($r = 0.456, p < 10^{-15}$). Blue points denote intervals where the Rank 1 alter was retained; red points denote intervals where the Rank 1 alter was replaced.

5.3 Multilevel Panel Models of Signature Divergence

To identify the structural and psychological drivers of signature mutation over time, we estimated linear mixed-effects panel models with ego random intercepts:

$$\text{JSD}_{it} = \beta_0 + \beta_1 \text{Turnover}_{it} + \beta_2 \Delta\text{Activity}_{it} + \mathbf{X}_{it}\gamma + u_i + \epsilon_{it} \tag{16}$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ and $\epsilon_{it} \sim \mathcal{N}(0, \sigma_\epsilon^2)$.

As reported in Table 5, alter turnover is the primary driver of signature divergence ($\beta = 0.351, p < 10^{-15}$), followed by large shifts in communication volume ($\beta = 0.0014, p < 10^{-10}$). Net of turnover, egonet clustering does not significantly alter signature stability, underscoring that signatures reflect cognitive rather than purely topological constraints.

Table 5: Multilevel Linear Mixed-Effects Models Predicting Signature Self-Divergence (JSD)

Predictor Term	**Model 1 (Turnover)**	**Model 2 (Topology)**	**Model 3 (Integrated)**
Intercept	-0.237^{***} (0.0195)	-0.200^{***} (0.0224)	-0.171^{***} (0.0320)
Alter Turnover ($1-$ Jaccard)	0.373^{***} (0.0242)	0.336^{***} (0.0240)	0.351^{***} (0.0287)
% Activity Change ($\|\Delta$Calls$\|$/Calls)	–	0.0014^{***} (0.0002)	0.0014^{***} (0.0002)
Egonet Clustering Coefficient	–	-0.0145 (0.0157)	-0.0066 (0.0153)
Egonet Density	–	0.0029 (0.0154)	–
Baseline Extraversion	–	–	$-0.0069^{\dagger}$ (0.0038)
Baseline Neuroticism	–	–	$-0.0090^{\dagger}$ (0.0049)
Baseline CES-D Depression	–	–	0.0002 (0.0005)
Observations	870	870	870
Ego Groups	290	290	290

Standard errors in parentheses. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.

5.4 Personality Determinants of Signature Steepness

What psychological dispositions generate steep (core-concentrated) versus flat (diffuse) social signatures? Regressing ego-mean power-law decay exponents (α_i) on Big Five personality traits and personal network degree (Table 6 and Figure 7) reveals:

Table 6: Personality Determinants of Social Signature Power-Law Alpha (α_i)

Term	**Estimate**	**Std. Error**	**t value**	**p-value**
(Intercept)	0.680^{***}	0.162	4.21	< 0.001
Extraversion	-0.017	0.017	-1.01	0.314
Neuroticism	0.098^{***}	0.019	5.08	< 0.001
Agreeableness	0.063^{*}	0.025	2.49	0.014
Conscientiousness	0.021	0.023	0.89	0.374
Openness	0.014	0.027	0.53	0.598
Personal Network Degree	$-0.004^{\dagger}$	0.002	-1.76	0.080

$N = 290$ egos, $R^2 = 0.124, F(6, 283) = 6.69, p < 0.000001$.

Neuroticism is a highly significant, positive predictor of signature steepness ($\beta = 0.098, t = 5.08, p < 10^{-6}$). Individuals high in neuroticism exhibit hyper-concentrated communication profiles, funneling the vast majority of their interactions into one or two primary alters while under-investing in intermediate and outer bands. Agreeableness also exhibits a modest positive association with concentration ($\beta = 0.063, p = 0.014$). In contrast, extraversion tends to flatten signatures toward broader alter engagement.

6 Discussion and Conclusion

By linking continuous passive smartphone logs with multi-wave sociocentric surveys and psychological batteries in the NetHealth cohort, this study provides definitive empirical backing for social signature theory while advancing the paradigm into sociological and psychological domains.

Our key findings are threefold:

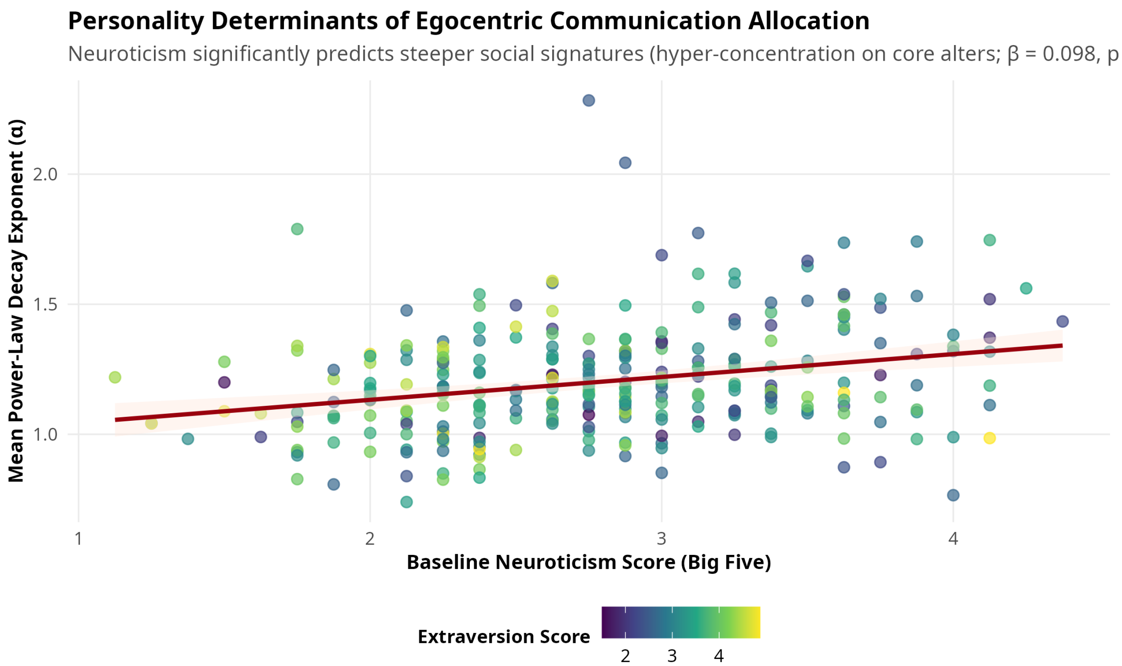

Figure 7: Personality Predictors of Social Signature Alpha. Neuroticism score predicts significantly steeper power-law decay ($\beta = 0.098, p < 10^{-6}$). Points colored by individual Extraversion score.

1. **Unambiguous Persistence**: Social signatures in NetHealth are consistently persistent across timescales ranging from 3-week rolling windows to multi-year academic terms. An individual's signature divergence across time is less than half the divergence observed between random peers.

2. **Substantive Grounding**: Social signature ranks are not arbitrary artifacts of call logging; they map directly to evolutionary Dunbar tiers. Rank 1 represents a specialized kin-based support hub, Ranks 2–5 represent the core emotional confidant circle, and ranks beyond 5 represent peer-based companionship.

3. **The Psychological Signature**: The shape of the social signature is anchored in personality. Neuroticism drives individuals to adopt steep, risk-averse, hyper-concentrated allocation strategies, while agreeable and emotionally stable individuals allocate attention across broader social circles.

Future research should leverage high-resolution wearable sensing (Fitbit physical activity, sleep regularity, and colocation traces) to examine whether disruptions in social signature equilibrium serve as early-warning biomarkers for loneliness, academic distress, and depressive episodes during major life transitions.

References

Gamal Adel, Shrichand Bhuria, Alessandro Catalano, Liber Dorizzi, Leonardo Federici, Theodora Moldovan, Berné Nortier, Chara Deanna Punzal, Giulia de Meijere, and Gerardo Iñiguez. Preferentiality and bandwidth drive tie activity in online and offline ego networks. *arXiv preprint arXiv:2606.27937*, 2026.

Simone Centellegher, Eduardo López, Jari Saramäki, and Bruno Lepri. Personality traits and ego-network dynamics. *PLOS ONE*, 12(3):e0173110, 2017.

Richard L. Daft and Robert H. Lengel. Information richness: A new approach to managerial behavior and organization design. *Research in Organizational Behavior*, 6:191–233, 1983.

Robin I.M. Dunbar. Neocortex size as a constraint on group size in primates. *Journal of Human Evolution*, 22(6):469–493, 1992.

Robin I.M. Dunbar. The social brain hypothesis. *Evolutionary Anthropology: Issues, News, and Reviews*, 6(5):178–190, 1998.

Robin I.M. Dunbar. The anatomy of friendship. *Trends in Cognitive Sciences*, 22(1):32–51, 2018.

Sara Heydari. *Dynamic yet Persistent: Investigating Digital Traces of Human Behaviour*. PhD thesis, Department of Computer Science, Aalto University School of Science, 2024. Aalto University publication series DOCTORAL THESES 118/2024.

Sara Heydari, Gerardo Iñiguez, János Kertész, and Jari Saramäki. Disentangling degree and tie strength heterogeneity in egocentric social networks. *EPJ Data Science*, 13(1):76, 2024.

Somayeh Heydari, Sam G.B. Roberts, Robin I.M. Dunbar, and Jari Saramäki. Multichannel social signatures and persistent features of ego networks. *Applied Network Science*, 3(1):8, 2018.

Gerardo Iñiguez, Sara Heydari, János Kertész, and Jari Saramäki. Universal patterns in egocentric communication networks. *Nature Communications*, 14(1):5064, 2023.

Weiwen Jia, Bin Zhou, Xin Lu, and Xiaoke Xu. Multidimensional social signature de-anonymizes low-sensitivity data. *Scientific Reports*, 15(1):16663, 2025.

Olessia Y. Koltsova, Larisa V. Mararitsa, Maxim A. Terpilovskii, and Yadviga E. Sinyavskaya. Social signature in an online environment: Stability and cognitive limits. *Computers in Human Behavior*, 122:106856, 2021.

Ren-De Li, Jian-Guo Liu, Qiang Guo, and Yi-Cheng Zhang. Social signature identification of dynamical social networks. *Physica A: Statistical Mechanics and its Applications*, 508:213–222, 2018.

Yue Li and Robert M. Bond. Evidence of the persistence and consistency of social signatures. *Applied Network Science*, 7(1):10, 2022.

Jianhua Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991.

Bo Liu, Xiao-Ke Xu, and Linyuan Lü. Uncovering patterns of multichannel mobile communications. *Physica A: Statistical Mechanics and its Applications*, 636:129546, 2024.

Giovanna Miritello, Esteban Moro, Rubén Lara, Rocío Martínez-López, and Petter Holme. Time as a limited resource: Communication strategy in mobile phone networks. *Social Networks*, 35(1): 89–95, 2013.

Sam G.B. Roberts, Robin I.M. Dunbar, Thomas V. Pollet, and Toon Kuppens. Exploring variation in active network size: Constraints and predictors. *Social Networks*, 31(2):138–146, 2009.

Jari Saramäki, Elizabeth A. Leicht, Eduardo López, Sam G.B. Roberts, Felix Reed-Tsochas, and Robin I.M. Dunbar. Persistence of social signatures in human communication. *Proceedings of the National Academy of Sciences*, 111(3):942–947, 2014.

Arkadiusz Stopczynski, Vedran Sekara, Piotr Sapiezynski, Andrea P. Zheng, Rolf Turlach, and Sune Lehmann. Measuring large-scale social networks with high resolution. *PLOS ONE*, 9(4):e95978, 2014.

Alistair Sutcliffe, Robin Dunbar, Jens Binder, and Holly Arrow. Relationships and the social brain: Integrating psychological and evolutionary perspectives. *British Journal of Psychology*, 103(2): 149–168, 2012.

Ignacio Tamarit, Angel Sánchez, and José A. Cuesta. Beyond Dunbar circles: A continuous description of social relationships and resource allocation. *Scientific Reports*, 12(1):2152, 2022.